

live RLB gains

$$\hat{d}^{\text{live}}, \hat{o}^{\text{live}}$$

matrix pressure

$$q^{\text{in}}, q^{\text{out}}, \pi$$

coefficient activity

$$q^R, \alpha$$

post-descent pair

$$\hat{d}^{\text{probe}}, \hat{o}^{\text{probe}}, \gamma$$

stop-gradient · detached read-only summary bus

1 Group-gradient
gate

$$\nabla A_{l,g}, \nabla B_{l,g} \mapsto \tilde{\nabla} A_{l,g}, \tilde{\nabla} B_{l,g}$$

reads live gains
 π, α

2 Role/depth
matrix step

AdamW role/depth
early Muon pulse
 $(A_l^{\text{desc}}, B_l^{\text{desc}})$
Muon damping π, α

3 Reciprocal
pair rescale

$$\begin{aligned} A_{l,g}^{t+1} &= e^{\ell} A_{l,g}^{\text{desc}} \\ B_{l,g}^{t+1} &= e^{-\ell} B_{l,g}^{\text{desc}} \end{aligned}$$

reads probes
 γ, π, α

updated pair
 A_l^{t+1}, B_l^{t+1}

clipped
gradients
 $\nabla A_l, \nabla B_l$

current pair
 A_l^t, B_l^t

rational coefficients and non-RLB tensors

AdamW

ordinary updates